

Emergency Operations Plan (EOP)

Chiller - Centrifugal Failure to Start

Section 01 - EOP Schedule Information

EOP Title	Chiller - Centrifugal Failure to Start
EOP Identifier	EOP-CH-303-FailureToStart
Version:	1
Creation Date	-
Work Description:	On failure to start from Daikin WME106CDCSNA Chiller CH-303, respond to and stabilize by addressing control and power issues, including alarm assessment, control system verification, and ensuring proper water flow.
Component Type:	Chiller - Centrifugal
Manufacturer:	Daikin
Model Number:	WME106CDCSNA
Serial Number:	STNU241100080
Equipment Number:	CH-303
Location	MCP-3

Duration	30-60 minutes (estimate)
Level of Risk (LOR):	medium
CET Level Required:	CET II
Author:	-
Author CET Level:	-
Approver	-

Section 02: Site Information

Customer	Liam Fleming
Site Name	LF001
Data Center Location:	Richmond Virginia
Site Address:	6920 Forest Ave
Site Contact:	undefined

Section 03: EOP Overview

EOP Title:	Chiller - Centrifugal Failure to Start
Work Area:	MCP-3
Building/Floor/Room:	MCP-3

Access Requirements:	Badge access to the work area required. Escort by site Facilities Operations for non-authorized personnel. PPE per facility safety policy. (Site Policy not attached; verify with facility access control.)
Self Delivered / Vendor:	self_delivered
Qualifications Required:	Manufacturer-authorized training on the Daikin WME106CDCSNA chiller required. Site emergency-response training and site safety orientation must be current.
Immediate notifications required:	Notify the on-shift supervisor, Operations or NOC, and Facilities Management immediately upon recognition of the event, via the site emergency communication plan.
Post notifications required:	Notify Operations and Facilities Management of stabilization and all-clear immediately upon recovery. Submit an incident report and root cause analysis to Site Management and the Safety Officer.

Section 04: Immediate Emergency Actions - Power Failure Diagnostics

Pre-Action Safety & Equipment Requirements:

Equipment-Specific PPE Requirements for CH-303

PPE Item	Specification for CH-303	Verified
Safety glasses with side shields; use chemical splash goggles plus face shield if refrigerant or oil release is suspected	ANSI/ISEA Z87.1 eye and face protection rated for impact and liquid splash	
Hand protection for external inspection, sharp edges, and possible refrigerant or POE oil contact	ANSI/ISEA 105 cut-resistant, liquid-resistant insulated gloves suitable for cold liquid refrigerant and oil contact	

PPE Item	Specification for CH-303	Verified
Long-sleeve protective work clothing covering arms and legs	OSHA 29 CFR 1910.132 protective clothing selected to prevent skin contact with refrigerant, oil, and sharp edges	
Safety footwear	ASTM F2413 protective-toe, slip-resistant safety footwear	
Pressure-demand self-contained breathing apparatus for suspected refrigerant release or oxygen-deficient atmosphere	NIOSH-approved SCBA certified under 42 CFR Part 84 for IDLH or oxygen-deficient atmospheres	
Hearing protection when mechanical-room noise is elevated	OSHA 29 CFR 1910.95 hearing protection with adequate NRR for the measured or expected noise level	

Required Tools & Test Equipment for CH-303

Tool/Equipment	Specific Model/Type for CH-303	Available
Daikin MicroTech HMI touch screen	OEM unit controller interface for alarms, start permissives, fault history, and setpoints	
BAS workstation or remote monitoring terminal	BAS interface for remote enable command, alarm status, and chilled-water plant sequencing	
Two-way radio	site communications for coordinating with MCP-3, control room, and electrical/mechanical responders	
Portable flashlight	intrinsically safe not required; suitable for mechanical room inspection	
True-RMS multimeter	CAT III 600V minimum, AC/DC voltage verification for control power or supply status when authorized	
Digital differential pressure meter	water-side pressure drop verification across evaporator/condenser P/T ports	
USB flash drive	USB data capture for HMI alarm or trend log export	

CRITICAL SAFETY CHECKPOINT - STOP Before Proceeding

Step Number	Safety Checklist Items
1	MCP-3 is safe to approach and enter, with no active smoke, fire, refrigerant alarm or odor, abnormal noise, flooding, or visible electrical distress at CH-303; any uncontrolled condition has been escalated and entry is withheld until controlled.
2	Only authorized and qualified responders familiar with centrifugal chiller emergency response are entering the CH-303 work area.
3	Required PPE for mechanical-room chiller response is available and donned, including eye protection, safety footwear, gloves appropriate for external inspection, and hearing protection if sound levels require it.
4	Two-way communication is established with operations or the control room, and chiller, mechanical, electrical, and vendor escalation contacts are available before starting the response.
5	Current BAS/HMI status, active alarms, alarm history, unit status, compressor status, and recent trend data for CH-303 have been reviewed before any reset or start attempt.
6	The impact of CH-303 being unavailable on chilled-water capacity, critical load cooling, and remaining redundancy is understood and communicated to operations before acting.
7	The initial response scope is limited to HMI/BAS review, external visual inspection, and operator-level reset or start commands; no electrical panels, compressor access covers, or other energized enclosures will be opened during this response.
8	No personnel are positioned near rotating equipment, couplings, belts, fans, valves, or automatic-start equipment that could move if CH-303 or associated pumps receive a start command.
9	Evaporator and condenser water systems serving CH-303 show no visible leaks, flooding, ruptured piping, or conditions indicating unsafe water release near the unit before the response begins.
10	Available HMI/BAS indications show evaporator and condenser water flow paths, pump status, and water temperatures are known before any start attempt is considered.

 **DO NOT PROCEED** until all safety requirements are verified for CH-303

Internal Equipment Diagnostics for CH-303

Perform systematic internal component checks to identify the source of Power Failure

Step Number	Internal Component to Check	Expected Reading/Condition	Actual Reading	Pass/Fail
1	HMI Home screen unit status and compressor status	With a valid start request, Unit Status advances to Starting or Running and compressor status advances through Pre-Start, Starting, then Running per MicroTech status possibilities; unit is not remaining Off or Shutdown.		
2	HMI ALARM Active screen	No active Critical Alarm shown in red and no active Warning shown in yellow that would shut down, inhibit, or limit compressor start. Manual-defined critical alarms shut the unit or compressor off if not corrected; warnings may limit loading or indicate an out-of-normal condition.		
3	HMI Alarm History and most recent action taken	Most recent alarm history does not show a repeated start-inhibit cause such as Evaporator Water Flow Loss, Condenser Water Flow Loss, Evaporator Pump Fault, Condenser Pump Fault, Low Evaporator Pressure, High Condenser Pressure, Evaporator Pressure Sensor Fault, or Condenser Pressure Sensor Fault.		
4	Unit enable source and stop/start permissive status	Selected enable source is allowing operation; Unit Status is not Off - Manual Switch, Off - HMI (Local), Off - Remote Switch, or Off - BAS Network. Front panel Stop/Auto and Remote Start/Stop digital input states indicate Auto where applicable.		
5	Cooling demand, leaving water temperature setpoint, and demand limit input	Controller shows a valid cooling demand with leaving evaporator water temperature above the active chilled-water setpoint, and the demand limit input is not restricting the unit to a no-start or zero-capacity condition.		
6	Evaporator pump output/status and evaporator water flow switch input	Evaporator pump control/status is available and the Evaporator Water Flow Switch digital input indicates Flow. Manual alarm logic: Evaporator Pump Fault occurs if flow is not seen for 60 seconds after the evaporator pump turns on when pump control is enabled.		

Step Number	Internal Component to Check	Expected Reading/Condition	Actual Reading	Pass/Fail
7	Evaporator water flow indication and evaporator pressure drop	Evaporator water flow and pressure drop are correct for the installed system and flow is stable; no Evaporator Water Flow Loss indication is present.		
8	Entering and leaving evaporator water temperatures	Entering and leaving evaporator water temperatures are valid and within the WME acceptable standby/operating range for the installed refrigerant and fluid; no freeze-protect or low-temperature inhibit is indicated.		
9	Condenser pump output/status and condenser water flow switch input	Condenser pump control/status is available and the Condenser Water Flow Switch digital input indicates Flow. Manual alarm logic: Condenser Pump Fault occurs if flow is not seen for 60 seconds after the condenser pump turns on when pump control is enabled.		
10	Condenser water flow indication and condenser pressure drop	Condenser water flow and pressure drop are correct for the installed system and flow is stable; no Condenser Water Flow Loss indication is present.		
11	Entering and leaving condenser water temperatures and cooling tower control indications	Entering and leaving condenser water temperatures are valid and within normal operating range per the controller; cooling tower bypass valve, tower VFD, and tower stage outputs show normal availability for condenser water temperature control when required.		
12	Evaporator pressure, suction pressure, and low evaporator pressure alarm status	Evaporator pressure and suction pressure readings are valid and above the Low Evaporator Pressure Alarm Setpoint; no Low Evaporator Pressure or Evaporator Pressure Sensor Fault is active. Manual-listed causes include low/no evaporator water flow, low refrigerant level in evaporator, or incorrect leaving water temperature setpoint.		

Step Number	Internal Component to Check	Expected Reading/Condition	Actual Reading	Pass/Fail
13	Condenser pressure, discharge pressure, and high condenser pressure alarm status	Condenser pressure and discharge pressure readings are valid and below the High Condenser Pressure setpoint; no High Condenser Pressure or Condenser Pressure Sensor Fault is active.		
14	Compressor detail screen state information for each compressor	Compressor state sequence proceeds normally through pre-start events without remaining in a hold, shutdown, pre-shutdown, or Power Fail Detected state. Manual compressor status possibilities include Off, Pre-Start, Starting, Running, Run IGV, Run Speed, Run Hold, Power Fail Detected, Pre-Shutdown, and Shutting Down.		
15	Compressor fault log and MicroTech last fault information	Fault log and last fault analysis show no active compressor start-inhibit fault and no recurring fault pattern associated with magnetic bearing, compressor controller, VFD, IGV, pressure, temperature, or flow permissives.		
16	IGV position/status and start permissive indication	Inlet guide vane status is valid, no IGV fault is present, and indicated IGV position matches the controller's commanded start/pre-start condition for the compressor.		
17	EXV driver signal and refrigerant control status	EXV driver signal is valid within the controller output range of 0 to 10 VDC corresponding to 0 to 100% open, and no expansion valve driver or refrigerant control fault is indicated that would inhibit start.		
18	Temperature and pressure sensor validity on MicroTech detail screens	Evaporator leaving water temperature, entering evaporator water temperature, entering/leaving condenser water temperature, evaporator pressure, condenser pressure, suction pressure, and discharge pressure values are present, reasonable, and not flagged bad or disconnected by the controller.		

Step Number	Internal Component to Check	Expected Reading/Condition	Actual Reading	Pass/Fail
19	Unit controller, HMI, and field I/O communication status	HMI communicates normally with the MicroTech controller, field I/O points display current states, and compressor controller communication is normal with no controller reset or communication fault indicated.		
20	Power tab and compressor power-fault indication	Power information is present and normal for the installed unit, and compressor status does not show Power Fail Detected. Verify this only after start request, alarms, permissives, flow, pressure, and controller status have been reviewed.		

⚠ WARNING: If internal diagnostics do not identify the cause of Power Failure, proceed to Section 05 for external equipment diagnostics. The fault may be upstream from the unit.

Section 05: Power Failure Detection External Response Actions

Verify all external equipment and systems that connect to or support the CH-303

Step Number	External Equipment/System to Check	Connection to CH-303	Potential Failure Mode Causing Power Failure	Verification Method	Actual Status	Pass/Fail
1	BAS or remote start/enable source serving CH-303	Provides the external start command or enable signal to the chiller controller through the BAS network or remote switch input	No start command present, remote enable open, BAS source commanding Off, or unit disabled from the external control source	Review BAS command/status and CH-303 HMI control source and unit status; confirm the external enable/start command is active and not commanding Off		

Step Number	External Equipment/System to Check	Connection to CH-303	Potential Failure Mode Causing Power Failure	Verification Method	Actual Status	Pass/Fail
2	Demand limit or load-shed signal serving CH-303	Provides an external 4-20 mA demand limit input that can restrict chiller loading or prevent start depending on command level and configuration	Demand limit signal active at a restrictive value or load-shed command preventing the chiller from starting	Review BAS demand-limit command and CH-303 HMI demand-limit indication; confirm the external signal is normal for start permissive conditions		
3	Chilled water system serving CH-303	Provides evaporator water flow required for the chiller start sequence and cooling function	Insufficient chilled water flow, isolation valve closed, loop not available, or evaporator entering water outside startup range	Review BAS chilled water loop status and local gauges/temperature indications; confirm flow is established and evaporator water conditions are within the documented startup range for the unit		
4	Evaporator pump system serving CH-303	External evaporator pump start/status circuit supports required chilled water flow through the evaporator	Evaporator pump unavailable, stopped, failed to prove status, or not responding to the chiller/BAS start sequence	Review BAS/pump controller status and local pump indication for the evaporator pump(s) serving CH-303; confirm a pump is running or available as required for start		

Step Number	External Equipment/System to Check	Connection to CH-303	Potential Failure Mode Causing Power Failure	Verification Method	Actual Status	Pass/Fail
5	External evaporator flow proof device or flow switch circuit	Provides flow proof permissive to the chiller controller for evaporator water flow	Flow proof not made due to inadequate flow, open proof circuit, or field device not indicating flow	Review CH-303 HMI flow permissive/status and BAS flow proof indication; compare with observed loop flow indication where available		
6	Condenser water system serving CH-303	Provides condenser water flow and heat rejection path required for chiller startup and operation	Insufficient condenser water flow, isolation valve closed, loop not available, or condenser water temperature outside startup range	Review BAS condenser water loop status and local gauges/temperature indications; confirm flow is established and condenser water conditions are within the documented startup range for the unit		
7	Condenser pump system serving CH-303	External condenser pump start/status circuit supports required water flow through the condenser	Condenser pump unavailable, stopped, failed to prove status, or not responding to the chiller/BAS start sequence	Review BAS/pump controller status and local pump indication for the condenser pump(s) serving CH-303; confirm a pump is running or available as required for start		

Step Number	External Equipment/System to Check	Connection to CH-303	Potential Failure Mode Causing Power Failure	Verification Method	Actual Status	Pass/Fail
8	External condenser flow proof device or flow switch circuit	Provides flow proof permissive to the chiller controller for condenser water flow	Flow proof not made due to inadequate flow, open proof circuit, or field device not indicating flow	Review CH-303 HMI flow permissive/status and BAS flow proof indication; compare with observed loop flow indication where available		
9	Cooling tower system serving CH-303	Rejects heat from the condenser water loop and supports condenser water temperature control through tower stages, VFD output, and bypass valve interface	Cooling tower unavailable, tower fans disabled, bypass valve not in normal automatic control, or condenser water temperature not suitable for chiller start	Review BAS cooling tower status, tower fan/VFD availability, bypass valve position, and condenser water temperature trend serving CH-303		
10	Customer-supplied main power source feeding CH-303	Provides the external line power required by the chiller power panel and compressor drives	Upstream source unavailable, upstream protective device open, phase loss, or abnormal voltage preventing chiller start	Review EPMS or upstream distribution indication and CH-303 HMI power status; confirm the source feeding the unit is available with no upstream trip or power-fail indication		

Step Number	External Equipment/System to Check	Connection to CH-303	Potential Failure Mode Causing Power Failure	Verification Method	Actual Status	Pass/Fail
11	Customer-supplied control power / remote control power source for CH-303	Provides external control power used by the chiller control panel, remote enable circuits, and field interface wiring where applicable	Control power unavailable or abnormal, causing the controller or external start/enable interfaces to remain inactive	Verify control power status from the HMI/controller indication and upstream control power source indication; confirm the chiller controller and field interface circuits are energized		

Section 06: Communication & Escalation Protocol

Escalation Matrix

Level	Title	Contact Name	Phone Number
Level 0			

Emergency Contacts

Service Type	Contact Name/Organization	Phone Number	Notes/Address
Emergency (Police / Fire / EMS)		911	Site listed in Henrico County, Virginia; district: Tuckahoe.
Fire / EMS (Non-Emergency)	Henrico County Division of Fire / Henrico County Department of Emergency Communications	(804) 501-5000	Henrico County Division of Fire mailing address: P.O. Box 90775, Henrico, VA 23273-0775; non-emergency dispatches police, fire, and rescue units 24/7.

Service Type	Contact Name/Organization	Phone Number	Notes/Address
Police (Non-Emergency)	Henrico County Police Division	(804) 501-5000	7721 E. Parham Rd., Henrico, VA 23294; West Station/main headquarters.
Nearest Hospital / Emergency Room	Henrico Doctors' Hospital	(804) 289-4600	1602 Skipwith Rd, Richmond, VA 23229; 24/7 emergency room. Main phone: (804) 289-4500.
Electric Utility Emergency	Dominion Energy Virginia	1-866-366-4357	Dominion Energy Virginia outage/emergency reporting for downed power lines and outages.
Gas Utility Emergency	Richmond Gas Works	(804) 646-4646	Natural gas service to City of Richmond, Henrico, and northern Chesterfield; call for gas odor/rotten egg smell.
Poison Control	VCU Health Virginia Poison Center	(800) 222-1222	P.O. Box 980522, Richmond, Virginia 23298-0522; 24/7 poison help. Local: (804) 828-9123.
Environmental / Spill Response	Virginia Department of Environmental Quality Pollution Response Program / Virginia Emergency Operations Center	804-750-8845	Virginia DEQ Pollution Response Program responds to pollution incidents impacting air, land, and water; significant imminent-threat incidents directed to Virginia Emergency Operations Center.
Site Operations	Site Operations Manager		
Security	Site Security Control Room		
Facilities Support	Facilities Duty Engineer		
Electrical On-Call	Electrical On-Call Team		
Mechanical On-Call	Mechanical On-Call Technician		

Service Type	Contact Name/Organization	Phone Number	Notes/Address
Environmental Health & Safety	Site EHS Manager		

✓ RESEARCHED: Contact information has been researched for the specific location at 6920 Forest Ave . Please verify current phone numbers before use as they may change over time.

Contact Name	-
Phone Number	-

Section 07: Recovery & Return to Service

Power Failure Resolution and Equipment Recovery Procedures

Follow these steps in sequence to safely restore the CH-303 to normal operation after Power Failure has been resolved:

Power Failure Resolution Verification

Confirm stable operating conditions are available at all system levels:

- Verify the CH-303 area is clear of personnel, tools, temporary hoses, test equipment, and loose materials before enabling another start attempt.
- If lockout/tagout was applied during corrective work, verify only authorized employees have removed their locks/tags and all affected employees have been notified per site LOTO policy.
- If electrical panels, compressor access covers, guards, or control enclosures were opened during corrective work, verify they are reinstalled, secured, and free of tools or debris before the chiller is enabled.
- If compressor or VFD access covers were opened after power isolation, verify the required stored-energy precautions were observed, including the Daikin requirement to wait 20 minutes after disconnecting power before opening compressor access covers, and verify all temporary grounds or test leads have been removed.

- Verify no residual electrical hazard remains from the failed-start event or corrective work: no signs of overheating, arcing, damaged conductors, abnormal odor, or loose electrical components in accessible chiller, starter, and control sections.
- Verify no refrigerant hazard is present around the chiller: no indication of refrigerant leakage, relief discharge, abnormal pressure condition, or impaired mechanical-room ventilation.
- Verify evaporator and condenser water piping affected by the troubleshooting work is intact, leak-free, filled, vented, and in the normal start configuration.
- Verify chilled-water isolation valves, condenser-water isolation valves, bypass valves, and control valves associated with CH-303 are in their normal automatic/start positions for chiller operation.
- Verify evaporator and condenser water strainers are installed and clear enough to support required flow if they were inspected or cleaned during corrective work.
- Verify evaporator and condenser pumps associated with CH-303 are available for automatic operation, with starters/interlocks restored and no local safeties bypassed.
- Verify cooling tower fans, condenser-water temperature controls, and related tower/bypass controls are available in their normal automatic configuration for chiller start.
- Verify the evaporator flow switch and condenser flow proving devices are installed, wired, calibrated/returned to service, and not bypassed.
- Verify the chilled-water and condenser-water systems are treated and within Daikin startup operating limits for the WME-C chiller before another start is permitted.
- Verify equipment-room temperature and ventilation are within Daikin startup operating limits for the WME-C chiller.
- Verify CH-303 MicroTech controls, onboard interlocks, remote enable/BAS enable source, pump interlocks, and alarm safeties are in their normal enabled configuration for a controlled start, with no permissive intentionally defeated.
- Verify prior failed-start alarms or safety trips have been reviewed and reset only after the associated unsafe condition has been cleared, and that no active lockout, inhibit, or manual override prevents a safe start.

- Verify compressor magnetic-bearing and VFD-related protective functions are available and not bypassed, including shaft-position/levitation permissives required by the Daikin startup sequence.
- Verify the electronic expansion valve and refrigerant-side controls are available for automatic operation and not mechanically or electrically disabled from the failed-start troubleshooting.
- If CH-303 was de-energized during corrective work, restore unit control power and compressor power through the normal upstream disconnects before starting.
- If control power was lost, wait for the MicroTech controllers and HMI to complete their normal reboot and show the chiller available.
- Reset the cleared failure-to-start alarm or start-inhibit condition at the MicroTech controller.
- Set the chiller Software Enable selection to Enabled.
- Select the intended chiller control source, either local HMI or BAS remote control.
- Place CH-303 in the normal cooling start mode from the selected control source.
- Set the evaporator pump selection and condenser pump selection to the normal automatic duty mode for this start.
- Enable the cooling tower controls and condenser-water temperature controls to respond to the chiller start sequence.
- Issue the start command for CH-303 from the selected control source.
- Observe the controller command the evaporator pump to start and prove evaporator water flow and chiller load.
- Observe the controller begin the lead compressor start sequence after evaporator flow and load are proven.
- Observe the controller activate the magnetic bearings and confirm correct compressor shaft position.
- Observe the controller command the condenser pump to start, prove condenser water flow, and confirm the EXV is open.

- Observe the compressor start, reach calculated Start RPM and IGV position, and begin ramping to match cooling load.
- Gradually restore chilled-water load through the BAS or plant controls as CH-303 stabilizes on leaving-water-temperature control.
- Confirm CH-303 remains running in normal cooling operation with no active start inhibit before handing the unit to performance monitoring.
- During the initial stabilization period under connected cooling load, CH-303 remains in Unit Status "Running" with the compressor in a normal running state, with no recurrence of the original Failure to Start condition, no return to Pre-Start/Starting without completing, and no protective Shutdown/Off transition.
- The start-permissive conditions implicated in a centrifugal chiller non-start remain satisfied during sustained operation, including established evaporator water flow, confirmed condenser water flow, verified magnetic-bearing shaft levitation/position, and compressor operation after reaching its commanded start state.
- Actual leaving evaporator water temperature remains controlled to the active chilled-water setpoint under the restored load, with entering and leaving evaporator water temperatures stable and within the unit's documented operating range for the installed fluid/refrigerant application.
- Evaporator water flow and pressure drop remain stable and adequate, with no recurrence of low/no evaporator water flow indications and no Low Evaporator Pressure critical alarm or related warning during the stabilization period.
- Condenser water flow, condenser pump status, and condenser entering/leaving water temperatures remain stable, with no recurrence of condenser-flow-related start inhibition and no High Condenser Pressure critical alarm during sustained operation.
- Compressor speed, inlet guide vane position, compressor kW, and motor current output modulate smoothly with load demand, without hunting, short-cycling, repeated unloading to limits, or abnormal load limiting.
- Cooling tower control remains stable for the current operating mode, with tower bypass valve and fan/VFD staging responding normally and without rapid cycling that destabilizes condenser water temperature or condenser pressure.

- Machine-health indications remain normal for the Daikin WME magnetic-bearing chiller, including evaporator/condenser pressure readings, suction/discharge indications, expansion valve operation, and no evaporator or condenser pressure sensor fault alarms.
- The HMI shows no active critical alarms or warnings during sustained operation, and the alarm history since restart shows no repeat of the original non-start alarm/condition and no new protective shutdown or compressor loading restriction.
- HMI trend data since restart shows stable evaporator temperatures, condenser temperatures, percent compressor speed, and compressor kW, confirming the chiller remained online and carried load without another failed-start attempt or shutdown cycle.
- Return CH-303 and its associated chilled-water, condenser-water, and cooling-tower support equipment to the normal automatic control mode and normal control source used for routine plant operation.
- Remove any manual commands, forced points, overrides, test modes, or temporary operator holds introduced during the CH-303 failure-to-start response, including chiller enable/start commands, remote enable, demand limit, leaving-water reset, evaporator pump, condenser pump, tower stage, and tower bypass valve points; restore normal setpoints.
- Reinstate normal chiller plant rotation, lead/lag sequence, staging, dispatch, and CH-303 availability status if these were changed while responding to the failure-to-start incident.
- Remove and document removal of any temporary bypasses, jumpers, inhibited interlocks, or isolations applied to start permissives, pump interlocks, flow switches, safeties, or control circuits during the failure-to-start response.
- Restore CH-303 monitoring and alarm configuration to normal, including re-enabling any suppressed, inhibited, or temporarily rerouted alarm points associated with the failure-to-start event and supporting equipment.
- Archive the alarm and event history for the CH-303 failure-to-start incident, including HMI alarm history, HMI trend logs, compressor fault log, and last-fault analysis; attach the records to the incident and work-order files. If manufacturer support is engaged, retain or transmit the required Trend.csv and AlarmHistory.csv files together as specified in the manual.
- Complete the failure-to-start incident record with the event description, timeline, operational impact, confirmed root cause, corrective actions taken, temporary measures used, and final return-to-normal time.

- Update and close the site CMMS/work-order record in accordance with site work-order policy, cross-referencing asset CH-303, the failure-to-start incident record, the root-cause record, parts or consumables used, and any assigned follow-up actions.
- If lockout/tagout or energy isolation was applied during the response, have authorized employees remove devices only when ready for normal operation, notify affected employees when devices are removed, and record completion in accordance with the Glenart Group LOTO Policy.
- Remove any temporary cooling, load-management, alternate dispatch, or stakeholder communication measures introduced because CH-303 failed to start, once normal plant configuration has been restored and approved by operations.
- Notify operations, facility management, and affected stakeholders that the CH-303 failure-to-start incident is resolved and the chiller has been returned to normal operating configuration; use the site escalation and contact structure defined elsewhere in this EOP.
- Document shift turnover notes covering the CH-303 failure-to-start cause, resolution, restored normal configuration, archived alarm/history records, and any open follow-up items or operating restrictions.
- Capture lessons learned and recurrence-prevention actions from the failure-to-start root-cause review, including any improvements related to start permissives, remote enable, pump or tower interlocks, alarm response, or control setpoint management.
- Replenish any response consumables used and return temporary tools, test equipment, portable communications, access permissions, and response documentation packages to their normal storage or administrative status.

Confirm voltage readings at equipment disconnect:	VAC
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Pre-Start Safety Checks

Complete all safety verifications before energizing equipment:

- Verify the CH-303 area is clear of personnel, tools, temporary hoses, test equipment, and loose materials before enabling another start attempt.

- If lockout/tagout was applied during corrective work, verify only authorized employees have removed their locks/tags and all affected employees have been notified per site LOTO policy.
- If electrical panels, compressor access covers, guards, or control enclosures were opened during corrective work, verify they are reinstalled, secured, and free of tools or debris before the chiller is enabled.
- If compressor or VFD access covers were opened after power isolation, verify the required stored-energy precautions were observed, including the Daikin requirement to wait 20 minutes after disconnecting power before opening compressor access covers, and verify all temporary grounds or test leads have been removed.
- Verify no residual electrical hazard remains from the failed-start event or corrective work: no signs of overheating, arcing, damaged conductors, abnormal odor, or loose electrical components in accessible chiller, starter, and control sections.
- Verify no refrigerant hazard is present around the chiller: no indication of refrigerant leakage, relief discharge, abnormal pressure condition, or impaired mechanical-room ventilation.
- Verify evaporator and condenser water piping affected by the troubleshooting work is intact, leak-free, filled, vented, and in the normal start configuration.
- Verify chilled-water isolation valves, condenser-water isolation valves, bypass valves, and control valves associated with CH-303 are in their normal automatic/start positions for chiller operation.
- Verify evaporator and condenser water strainers are installed and clear enough to support required flow if they were inspected or cleaned during corrective work.
- Verify evaporator and condenser pumps associated with CH-303 are available for automatic operation, with starters/interlocks restored and no local safeties bypassed.
- Verify cooling tower fans, condenser-water temperature controls, and related tower/bypass controls are available in their normal automatic configuration for chiller start.
- Verify the evaporator flow switch and condenser flow proving devices are installed, wired, calibrated/returned to service, and not bypassed.
- Verify the chilled-water and condenser-water systems are treated and within Daikin startup operating limits for the WME-C chiller before another start is permitted.

- Verify equipment-room temperature and ventilation are within Daikin startup operating limits for the WME-C chiller.
- Verify CH-303 MicroTech controls, onboard interlocks, remote enable/BAS enable source, pump interlocks, and alarm safeties are in their normal enabled configuration for a controlled start, with no permissive intentionally defeated.
- Verify prior failed-start alarms or safety trips have been reviewed and reset only after the associated unsafe condition has been cleared, and that no active lockout, inhibit, or manual override prevents a safe start.
- Verify compressor magnetic-bearing and VFD-related protective functions are available and not bypassed, including shaft-position/levitation permissives required by the Daikin startup sequence.
- Verify the electronic expansion valve and refrigerant-side controls are available for automatic operation and not mechanically or electrically disabled from the failed-start troubleshooting.

Equipment-Specific Restart Sequence

Follow the manufacturer-specific startup procedure for CH-303

- If CH-303 was de-energized during corrective work, restore unit control power and compressor power through the normal upstream disconnects before starting.
- If control power was lost, wait for the MicroTech controllers and HMI to complete their normal reboot and show the chiller available.
- Reset the cleared failure-to-start alarm or start-inhibit condition at the MicroTech controller.
- Set the chiller Software Enable selection to Enabled.
- Select the intended chiller control source, either local HMI or BAS remote control.
- Place CH-303 in the normal cooling start mode from the selected control source.
- Set the evaporator pump selection and condenser pump selection to the normal automatic duty mode for this start.

- Enable the cooling tower controls and condenser-water temperature controls to respond to the chiller start sequence.
- Issue the start command for CH-303 from the selected control source.
- Observe the controller command the evaporator pump to start and prove evaporator water flow and chiller load.
- Observe the controller begin the lead compressor start sequence after evaporator flow and load are proven.
- Observe the controller activate the magnetic bearings and confirm correct compressor shaft position.
- Observe the controller command the condenser pump to start, prove condenser water flow, and confirm the EXV is open.
- Observe the compressor start, reach calculated Start RPM and IGV position, and begin ramping to match cooling load.
- Gradually restore chilled-water load through the BAS or plant controls as CH-303 stabilizes on leaving-water-temperature control.
- Confirm CH-303 remains running in normal cooling operation with no active start inhibit before handing the unit to performance monitoring.

Record startup time:

-

System Functionality Verification

Monitor critical parameters during the startup phase:

Parameter	Expected Range	Actual Reading	Pass/Fail
HMI Unit Status	Progresses from Starting to Running; not Shutdown or Off per documented Unit Status possibilities.		

Parameter	Expected Range	Actual Reading	Pass/Fail
Compressor Status for enabled compressor(s)	Progresses through Pre-Start/Starting to Running, Run IGV, Run Speed, or Run IGV/Speed as load is accepted; not Power Fail Detected, Pre-Shutdown, or Shutting Down.		
Compressor startup sequence requirements	Compressor Details startup sequence requirements show satisfied/green as the start sequence advances; no unsatisfied start permissive displayed.		
Evaporator pump and chilled-water flow proof	Evaporator pump status running as commanded and evaporator water flow confirmed by the controller; no low/no evaporator water flow indication.		
Magnetic bearing shaft levitation / shaft position	Magnetic bearings activated and correct shaft position verified before compressor start.		
Condenser pump and condenser-water flow proof	Condenser pump status running as commanded and condenser water flow confirmed by the controller during startup.		
Electronic expansion valve status / position	EXV open for startup permissive and modulating normally after compressor start.		
Compressor speed	Compressor reaches calculated Start RPM, then percent compressor speed increases or stabilizes as required to match cooling load.		
Inlet guide vane position / loading response	IGV position opens and modulates with compressor speed to match cooling load; stable and not held by an abnormal limit condition.		
Leaving evaporator water temperature versus active chilled-water setpoint	Actual leaving water temperature is at or approaching the active chilled-water setpoint; evaporator water remains within documented standby operating limits for the unit fluid/refrigerant configuration.		

Parameter	Expected Range	Actual Reading	Pass/Fail
Entering and leaving evaporator water temperature trend	Entering and leaving evaporator water temperatures display a normal cooling trend after compressor start, with leaving temperature lower than entering temperature under load.		
Evaporator water temperature operating/standby limit	For water: 38°F to 115°F for R-134a/R-515B units, or 38°F to 112°F for R-513A units. For water with antifreeze: 30°F to 115°F for R-134a/R-515B units, or 30°F to 112°F for R-513A units.		
Entering and leaving condenser water temperature trend	Entering and leaving condenser water temperatures display normal heat rejection after compressor start, with leaving condenser water warmer than entering condenser water under load.		
Evaporator refrigerant pressure	Stable and above the active Low Evaporator Pressure alarm setpoint; no evaporator pressure or suction pressure sensor fault indicated.		
Condenser refrigerant pressure	Stable and below the active High Condenser Pressure setpoint; no condenser pressure or discharge pressure sensor fault indicated.		
Compressor motor current / percent RLA	Within active amp limit, demand limit, or power limitation; unit motor current output scale is documented as 0–125% RLA.		
Compressor kW	Rises smoothly with compressor loading and remains stable below any active power limitation.		
Control source and unit enable status	Control source matches the intended operating source and Unit Enable is AUTO for normal automatic chiller operation.		
Active alarms / HMI alarm icon	No active Critical Alarm or Warning affecting startup or compressor loading; ALARM icon not flashing red or yellow.		

Load Transfer (if applicable)

This section is not typically applicable to a single Air Handling Unit.

Performance Validation

Confirm equipment is operating within normal parameters:

- During the initial stabilization period under connected cooling load, CH-303 remains in Unit Status "Running" with the compressor in a normal running state, with no recurrence of the original Failure to Start condition, no return to Pre-Start/Starting without completing, and no protective Shutdown/Off transition.
- The start-permissive conditions implicated in a centrifugal chiller non-start remain satisfied during sustained operation, including established evaporator water flow, confirmed condenser water flow, verified magnetic-bearing shaft levitation/position, and compressor operation after reaching its commanded start state.
- Actual leaving evaporator water temperature remains controlled to the active chilled-water setpoint under the restored load, with entering and leaving evaporator water temperatures stable and within the unit's documented operating range for the installed fluid/refrigerant application.
- Evaporator water flow and pressure drop remain stable and adequate, with no recurrence of low/no evaporator water flow indications and no Low Evaporator Pressure critical alarm or related warning during the stabilization period.
- Condenser water flow, condenser pump status, and condenser entering/leaving water temperatures remain stable, with no recurrence of condenser-flow-related start inhibition and no High Condenser Pressure critical alarm during sustained operation.
- Compressor speed, inlet guide vane position, compressor kW, and motor current output modulate smoothly with load demand, without hunting, short-cycling, repeated unloading to limits, or abnormal load limiting.
- Cooling tower control remains stable for the current operating mode, with tower bypass valve and fan/VFD staging responding normally and without rapid cycling that destabilizes condenser water temperature or condenser pressure.
- Machine-health indications remain normal for the Daikin WME magnetic-bearing chiller, including evaporator/condenser pressure readings, suction/discharge indications, expansion valve operation, and no evaporator or condenser pressure sensor fault alarms.

- The HMI shows no active critical alarms or warnings during sustained operation, and the alarm history since restart shows no repeat of the original non-start alarm/condition and no new protective shutdown or compressor loading restriction.
- HMI trend data since restart shows stable evaporator temperatures, condenser temperatures, percent compressor speed, and compressor kW, confirming the chiller remained online and carried load without another failed-start attempt or shutdown cycle.

Return to Normal Operation

Complete recovery documentation and notifications per Element Critical Limble CMMS Policy:

- Return CH-303 and its associated chilled-water, condenser-water, and cooling-tower support equipment to the normal automatic control mode and normal control source used for routine plant operation.
- Remove any manual commands, forced points, overrides, test modes, or temporary operator holds introduced during the CH-303 failure-to-start response, including chiller enable/start commands, remote enable, demand limit, leaving-water reset, evaporator pump, condenser pump, tower stage, and tower bypass valve points; restore normal setpoints.
- Reinstate normal chiller plant rotation, lead/lag sequence, staging, dispatch, and CH-303 availability status if these were changed while responding to the failure-to-start incident.
- Remove and document removal of any temporary bypasses, jumpers, inhibited interlocks, or isolations applied to start permissives, pump interlocks, flow switches, safeties, or control circuits during the failure-to-start response.
- Restore CH-303 monitoring and alarm configuration to normal, including re-enabling any suppressed, inhibited, or temporarily rerouted alarm points associated with the failure-to-start event and supporting equipment.
- Archive the alarm and event history for the CH-303 failure-to-start incident, including HMI alarm history, HMI trend logs, compressor fault log, and last-fault analysis; attach the records to the incident and work-order files. If manufacturer support is engaged, retain or transmit the required Trend.csv and AlarmHistory.csv files together as specified in the manual.
- Complete the failure-to-start incident record with the event description, timeline, operational impact, confirmed root cause, corrective actions taken, temporary measures used, and final return-to-normal time.

- Update and close the site CMMS/work-order record in accordance with site work-order policy, cross-referencing asset CH-303, the failure-to-start incident record, the root-cause record, parts or consumables used, and any assigned follow-up actions.
- If lockout/tagout or energy isolation was applied during the response, have authorized employees remove devices only when ready for normal operation, notify affected employees when devices are removed, and record completion in accordance with the Glenart Group LOTO Policy.
- Remove any temporary cooling, load-management, alternate dispatch, or stakeholder communication measures introduced because CH-303 failed to start, once normal plant configuration has been restored and approved by operations.
- Notify operations, facility management, and affected stakeholders that the CH-303 failure-to-start incident is resolved and the chiller has been returned to normal operating configuration; use the site escalation and contact structure defined elsewhere in this EOP.
- Document shift turnover notes covering the CH-303 failure-to-start cause, resolution, restored normal configuration, archived alarm/history records, and any open follow-up items or operating restrictions.
- Capture lessons learned and recurrence-prevention actions from the failure-to-start root-cause review, including any improvements related to start permissives, remote enable, pump or tower interlocks, alarm response, or control setpoint management.
- Replenish any response consumables used and return temporary tools, test equipment, portable communications, access permissions, and response documentation packages to their normal storage or administrative status.

Restoration completed by:	-	Restoration completed at:	-
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Section 08: References and Documentation

Company Policy Documents Consulted

The following company emergency response policies were referenced during EOP creation. The specific requirements from these policies have been incorporated directly into the emergency procedure steps in Sections 04 and 05 above.

Power Failure Resolution Verification

Confirm stable operating conditions are available at all system levels:

Policy Document	Upload Date	Type

Note: The emergency procedures in this EOP have been informed by the company policies listed above. All policy requirements have been expanded inline within the emergency action steps for immediate reference during emergencies.

Critical Infrastructure Locations

Infrastructure Element	Location Details	Access Requirements

Spare Parts Inventory

Critical spare parts for CH-303 emergency response:

Part Description	Part Number	Quantity	Storage Location

Related Documents

Section 09: EOP Approval & Review

Role	Name	Signature	Date
Author			
Approver			
Annual Review			